Instruments & Systems
Valve Automation & Control Technology

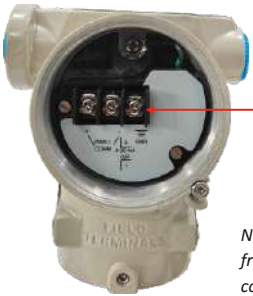
OPERATING MANUAL *for* TRANSMITTERS



1. General Information

Pressure Transmitter (ISPT-200) (Standard)





Note :- To reach out for menu buttons loosen the front cap of transmitter and for the electrical connection loosen the back side of the transmitter.

1. General Information

Pressure Transmitter with Customizable Electrical & Process Connection (ISPT-200)



Electrical Connection
Plug & Socket Type

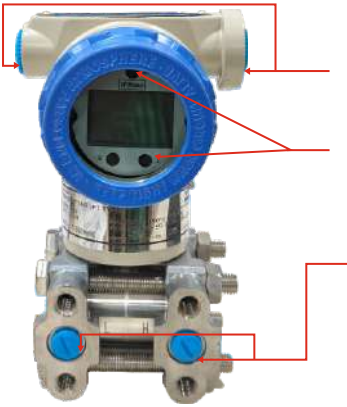
Process Connection with
Vent & Drain Plug



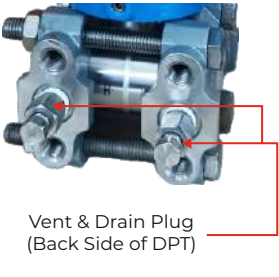
Electrical Connection
Harting Han Type

1. General Information

(i) Differential Pressure Transmitter (ISDPT-200) (Standard)



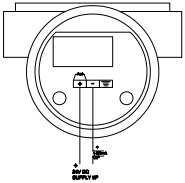
(ii) DPT (ISDPT-200) (Customizable Electrical & Process Connection)

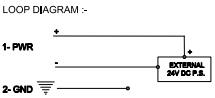




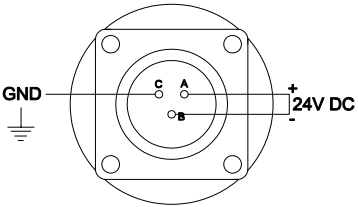
2. Electrical Connection

(a) Standard Electrical Connection





(b) Plug & Socket Type (3 Pin)



MALE PART

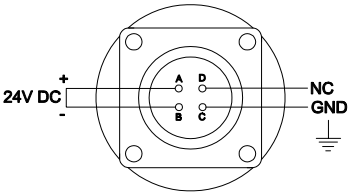
LOOP DIAGRAM:-

A - +

B - - **24V DC**

C - GND

(c) Plug & Socket Type (4 Pin)



MALE PART

LOOP DIAGRAM:-

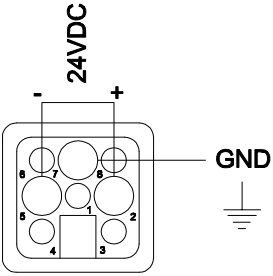
A - +

B - - **24V DC**

C - GND

D- NC (NO CONNECTION)

(d) Harting Han Connector

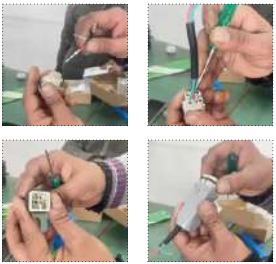


LOOP DIAGRAM:-

2. +
5. -
7. GND

24 VDC

Steps for wiring in Female Part of Harting Han Connector



(i) Crimp the un crimped pins with the wires as shown in the images.

(ii) Insert the crimped wires into the female part of the harting han connector so it passes through out the hole as shown in image.

(iii) Lock the female connector into the enclosure and tighten the screw as shown in the images.

(iv) Place the female part in front of male part and lower down the clip to lock it.



3. Mechanical Installation

(a) For Differential Pressure Transmitter (Accessories)



U Clamp



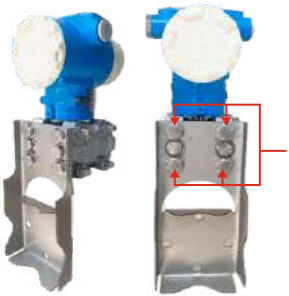
Washer & Nuts



Mounting Bracket

(a) For Differential Pressure Transmitter (Vertical Mounting)

STEP 1



(i) First you need to fix the brackets with 4 fastners provided in the mounting bracket kit.

(ii) This mounting plate will be directly tighten up with the transmitter.

STEP 2



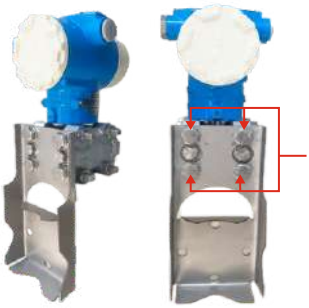
(i) Clamp the U Clamp from one side of the pipe and fix that mounting plate mounted with transmitter and pass it through the both horizontal holes of the mounting plate.

(ii) After that we have to fasten the nuts along with washers until it gets full tighten on pipe.

(iii) The mounting of transmitters on the vertical pipe will go like this. We can adjust/place the transmitter as per the site requirements.

(b) For Differential Pressure Transmitter (Horizontal Mounting)

STEP 1



(i) First you need to fix the brackets with 4 fastners provided in the mounting bracket kit.

(ii) This mounting plate will be directly tighten up with the transmitter.

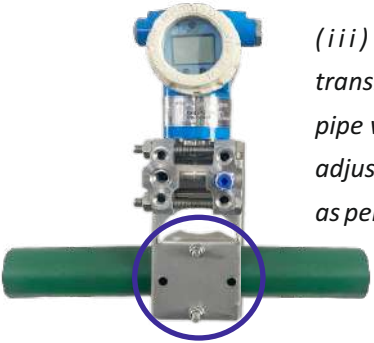
STEP 2



(i) Clamp the U Clamp from one side of the pipe and fix that mounting plate mounted with transmitter and pass it through the both vertical holes of the mounting plate.

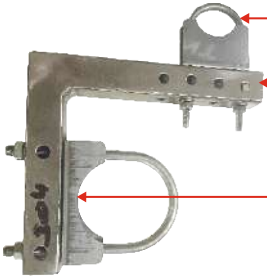


(ii) After that we have to fasten the nuts along with washers until it gets full tighten on pipe.



(iii) The mounting of transmitters on the vertical pipe will go like this. We can adjust/place the transmitter as per the site requirements.

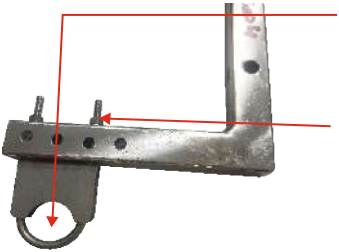
(c) For Pressure Transmitter (Vertical / Horizontal)





STEP 1

Place the C Clamp with the L Bracket then ahead of C Clamp put transmitter and fasten it with U Clamp using nuts & washers.



Note :

1. The smaller size U Clamp & C Clamp pair will be used for PT mounting.

2. The bigger size U Clamp & C Clamp pair will be used for Pipe/Panel.

3. The clamping should be done with L Bracket in vertical or horizontal holes as per the mounting pipe requirement.

STEP 2



1. Now clamp the other side of the bracket as the same.
2. Place the C Clamp with the L Bracket then ahead of C Clamp and tighten it with the pipe/panel with U Clamp using nuts & washers tightly.



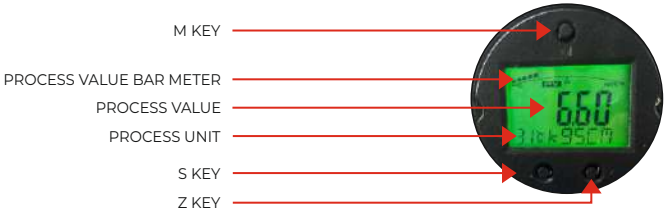
- The mounting arrangement of the PT in panel or pipe would be like this.
- This is the vertical pipe mounting it can be done with horizontal pipe mounting also, we just have to clamp the U Clamp in vertical holes of the L Bracket then the mounting will be as per horizontal pipe mounting.

Please note :-

1. for vertical pipe mounting the both U Clamp & C Clamp should be tighten in the horizontal holes.
2. for horizontal pipe mounting the both U Clamp & C Clamp should be tighten in the vertical holes.

4. Menu Buttons

Display & Function Keys



Adjustments with Keys

Menu 6: Zero Trim Setting

1. Key Z- Long press for 2-3 seconds. LCD will display a code- '00000'
2. Key Z- Again press this key for 4 times- '00000'
3. Key S- Press 6 times- '00006'
4. Key M- Press 1 time to go to Menu-6.
5. Key S- Press 2 times.
6. Key M- Press 2 times.
7. Key S- Lastly press 1 time to save the setting.

Menu 2: PV Unit Setting

8. Key Z- Long press for 2-3 seconds. LCD will display a code- '00000'
9. Key Z- Again press this key for 4 times- '00000'
10. Key S- Press 2 times- '00002'

11. Key M- Press 1 time to go to Menu-2.
12. Key S- Press 2 or more times to change the PV unit.
13. Key M- Press 6 times until it reaches the 0-Menu.
14. Key S- Lastly press 1 time to save the setting.

Menu 3 or 4: Zero / Span Range Setting

1. Key Z- Long press for 2-3 seconds. LCD will display a code- '00000'
2. Key Z- Again press this key for 4 times- '00000'
3. Key S- Press 3 times- '00003'
4. Key M- Press 1 time to go to Menu-3 and you may Press M 2 times to go to Menu-4.
5. Key S- Press 1 time (Arrow will Blink).
6. Key Z- Press 1 time (To go to the first digit).
7. Key S- Press 1 or more times (To change the digits value).
8. Key Z- Press 1 time (To go to the first digit).
9. Key Z- Press 6 times (To set the decimal place).
10. Key M- Press 5 times from Zero setting or 6 times from Span setting until it reaches the 0-Menu.
11. Key S- Lastly press 1 time to save the setting.

Menu 8: Output Type Setting

1. Key Z- Long press for 2-3 seconds. LCD will display a code- '00000'
2. Key Z- Again press this key for 4 times- '00000'
3. Key S- Press 8 times- '00008'
4. Key M- Press 1 time to go to Menu-8.
5. Key S- Press 2 times to alternates output as linear or square root.
6. Key M- Press 1 times.
7. Key S- Lastly press 1 time to save the setting.

5. Usage

Installing the transmitter after the power is switched on, it can be put into operation.
Each of the following must be checked before electrified :-
(i) Process connection (ii) Electrical connection
(iii) Guiding pipe of the transmitter and a measuring chamber are filled with dielectric.

Put into operation, follow the step operate cut-off valve.
Turn on the cut-off valve to connect with pressure;
Turn on the balance valve on the three-valves cluster (Differential pressure transmitter Only);
Turn on the cut-off valve of the higher pressure cavity;
Turn off the balance valve (Differential pressure transmitter Only);
Turn on the cut-off valve of the lower pressure cavity (Differential pressure transmitter Only);
For exiting operating transmitter, reverse steps.

4 ~ 20mA output signal Note: When measuring pressure is within the subject range of the nameplates, output current signal is in 4~20mA.
When the pressure is lower than the calibration of the measuring range, the output signal is between 3.9~4mA. When the pressure is higher than the calibration of the measuring range, the output signal is between 20~20.5mA.Output limit: Imin=3.9mA, Imax=20.5mA. Alarm current mode can be set. The low mode (min) is 3.7 mA. The high mode (max) is 21 mA. No-report mode (keep) is the effective value before the fault happening. Standard setting is the high mode.

Zero calibration
Because of zero offset caused by installation, the transmitter is needed to do digital zero adjustment.
The specific method for the next chapter: Menu Operations.
The control buttons of the transmitter are on the top of the transmitter. If needed, loosen the screw of the nameplates and rotate it 90 degree.

Zero and span calibration
To calibration smart transmitter using the following method:
1) Through operating control button;
2) Through communicator;
3) Through PC configuration software.
The first method is detailed in the "Menu Operations";

6. Maintenance & Repair

-Maintenance

According to the work conditions, inspecting on device's output reference should be at the previous section of periodic. Periodic inspection and cleaning are required according to the condition. Cleaning work would be the best to working indoor. Operation is as follows :-

1) Release the screws of process flange in accordance with the diagonal;

2) Be careful take down the flange and don't scratch on the sensor diaphragm;

3) Use soft brush and appropriate solvent to cleaning the sensor diaphragm and the flange;

4) Replacement process flange's O-rings;

5) Install process flange on the measuring unit, don't touch the isolating diaphragm (Both of the process flange's section must to be keep on the same plane, and hold on the right angle with the watch case);

6) Use the torque wrenches to diagonal for tighten in the bolts, nuts and check whether disclosure.

-Repair

The relevant safety requirements must be followed in the entire process of using. The transmitter could be disassembled on while in the cleaning, inspection, repairing and replacing of failed parts. The transmitter can only be repaired by the manufacturer. The failure of the transmitter should be returned to the manufacturer for repairing, if possible, please describe the problems and causes.